SCORES AI ANALYTICS

Systemic-Criticality Auto-Escalation Platform

A knowledge-graph system that reads every investor complaint, scores its systemic criticality, consolidates related cases, and auto-escalates the critical few to the right person at SEBI — with the evidence attached.

Product & Feature Specification

Prepared by OnFinance AI · Version 1.0

Builds on “Systemic-Criticality Auto-Escalation Framework” v1.0

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1 · Purpose & Scope

SEBI's SCORES portal already classifies investor complaints by regulated-entity type and grievance. That answers *"what kind of complaint is this?"* It does not answer the question that matters most to senior management: **"which of these, today, could signal harm to a large body of investors, a large quantum of money, or the integrity of the market — and therefore cannot sit in the routine queue?"**

The **Systemic-Criticality Auto-Escalation Framework (v1.0)** defined *what* should escalate. This document specifies the **product OnFinance AI has built to operationalise it**: an AI analytics layer that reads each complaint — its text, its regulated entity (RE), its workflow history and the surrounding population — assigns a **Systemic-Criticality Score (SCS)**, consolidates related complaints into single case-files, and **auto-escalates** the systemically significant ones to the appropriate SEBI officer with supporting evidence and reasoning.

1.1 What this product is, and is not

- **It is** a prioritisation, escalation and investigation layer that sits on top of SCORES. It surfaces and ranks; it attaches evidence; it notifies the right desk.
- **It is not** a replacement for SEBI's categories, nor an adjudication or enforcement system. Every escalation is a *recommendation* — explainable and reversible. SEBI officers decide.
- **It is fully isolated.** It reads the existing SCORES data **read-only** and writes nothing back; no existing SCORES service is modified (see §11).

1.2 What is new beyond the framework document

Beyond realising the 11-dimension scoring, this platform adds three capabilities the framework anticipated but did not specify: a **knowledge-graph** that makes systemic patterns visible; an **auto-escalation & notification engine** that pushes critical cases to designated officers (§9); and an **investigation surface** — an interactive graph, AI case summaries and a natural-language chatbot — so officers can act on an escalation in minutes.

2 · Product Overview — the four pillars

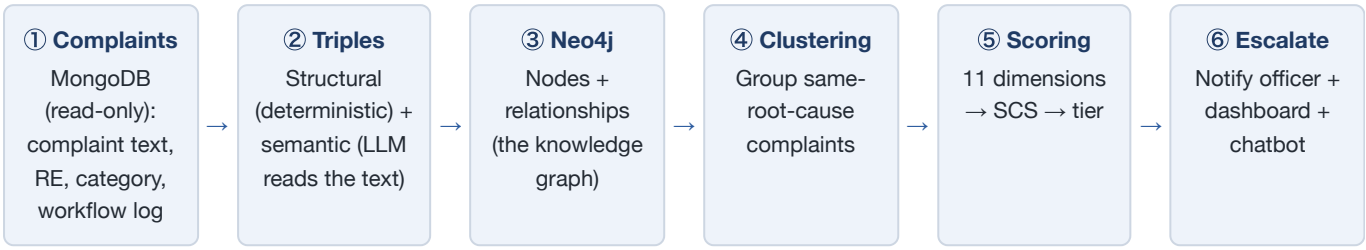
Pillar	What it does
1. Knowledge Graph	Every complaint, regulated entity, alleged wrongdoing (typology), product/scheme, investor and root-cause cluster becomes a connected node in a graph database (Neo4j). Patterns invisible in a table — 60 complaints about one IPO, a broker with a recurring fraud pattern, an issue spanning 8 states — become visible at a glance.
2. Criticality Scoring	Each complaint is assessed across **11 independent dimensions** (scale, money, fraud-severity, entity importance, velocity, vulnerability, market-integrity, distress, process-failure, cross-portal, legal). These combine into a single 0–100 **SCS** and a tier; any of nine **hard triggers** forces the top tier.
3. Auto-Escalation & Notification	When a case (or a consolidated cluster) crosses a tier or fires a hard trigger, the engine routes it to the **designated SEBI officer** — by email and an in-dashboard alert — with the score breakdown, evidence and a recommended SLA (§9).
4. Investigation surface	A live dashboard (ranked queue + knowledge-graph explorer), a one-line **AI summary** of every case, and an **Ask-the-Graph** chatbot that answers plain-English questions with a live, read-only query.

2.1 Current state (pilot snapshot)

Metric	Value	Note
Complaints in the graph	17,132	Full Q4-2024 SCORES population loaded structurally
AI-enriched & scored	~2,000	Validation slice; full-population enrichment is a background job
Regulated entities · timeline events	20,769 · 135,505	Graph totals: 183k+ nodes / 230k+ relationships
Root-cause & cross-RE clusters	615	Consolidated case-files (571 root-cause + 44 cross-RE product)
Tier distribution (scored slice)	320 Critical · 4 Priority · 143 Watch	320 critical complaints consolidate to ~21 Tier-3 case-files

3 · System Architecture & Data Pipeline

The platform is a **hybrid pipeline**: it uses a ruler where a fact is obvious and an AI where a fact is buried in free text. Nothing is written back to the SCORES database.



3.1 How complaint text becomes graph facts

A computer cannot reason over a paragraph, but it can reason over small three-part facts ("*subject* → *predicate* → *object*"). These are built two ways:

- **Deterministic facts (no AI).** "Which entity? Which category? Which state?" are already fields on the complaint. We copy them directly — instant, exact, no risk of an AI guessing when the answer is on the form.
- **AI-read facts (Claude Haiku 4.5).** "Is this fraud? How much money? Is the investor vulnerable?" live only inside the story. The AI reads each complaint under strict instruction — *report only what the text supports; never invent an amount or an allegation* — and returns tidy, typed signals.

> **Why hybrid.** The eleven dimensions are *defined* KPIs, so the AI's output is pinned to a fixed vocabulary (the ontology, §4). This keeps every signal both AI-powered *and* countable — you can ask "how many misappropriation cases?" and get a real number.

3.2 Isolation & safety

The entire platform runs as one standalone service. It consumes existing SCORES data through read-only reads and stores the graph and scores in its own database (Neo4j). **No existing SCORES, CRUD or front-end service is touched** (§11). The AI runs on Claude Haiku 4.5; for an air-gapped SEBI deployment it can be pointed at an on-prem LLM gateway without code changes.

4 · The Knowledge Graph & Ontology

The **ontology** is the fixed rulebook: exactly these *kinds of node* and exactly these *kinds of relationship*. That discipline is what makes the graph queryable and the KPIs countable.

4.1 Node types (the “pins”)

Node	Represents
Complaint	One investor grievance — carries the raw text, the structural fields, the AI signals, and the final score/tier
Entity	The regulated entity (broker, AMC, RTA, exchange…) complained against
Typology	The *kind* of alleged wrongdoing, from a Grave / High / Moderate library
Cluster	A consolidated “one root cause” case-file grouping many similar complaints
Product	The scheme / IPO / app / instrument named in the complaint
Investor	The complainant, stored as a one-way hash — no personal data is retained
Category · SubCategory · Location · TimelineEvent	SEBI taxonomy, geography, and the complaint’s workflow / ATR history

4.2 Relationship types (the “strings”)

Relationship	Meaning	Source
`FILED_AGAINST`, `IN_CATEGORY`, `LOCATED_IN`, `FILED_BY`, `HAS_EVENT`	which entity, category, state, investor, workflow log	Deterministic
`ALLEGES` → Typology, `IMPLICATES` → Product	the wrongdoing & product read from the text	AI
`SIMILAR_TO`, `MEMBER_OF`, `AGAINST`, `ON_PRODUCT`	semantic similarity & cluster membership	Derived

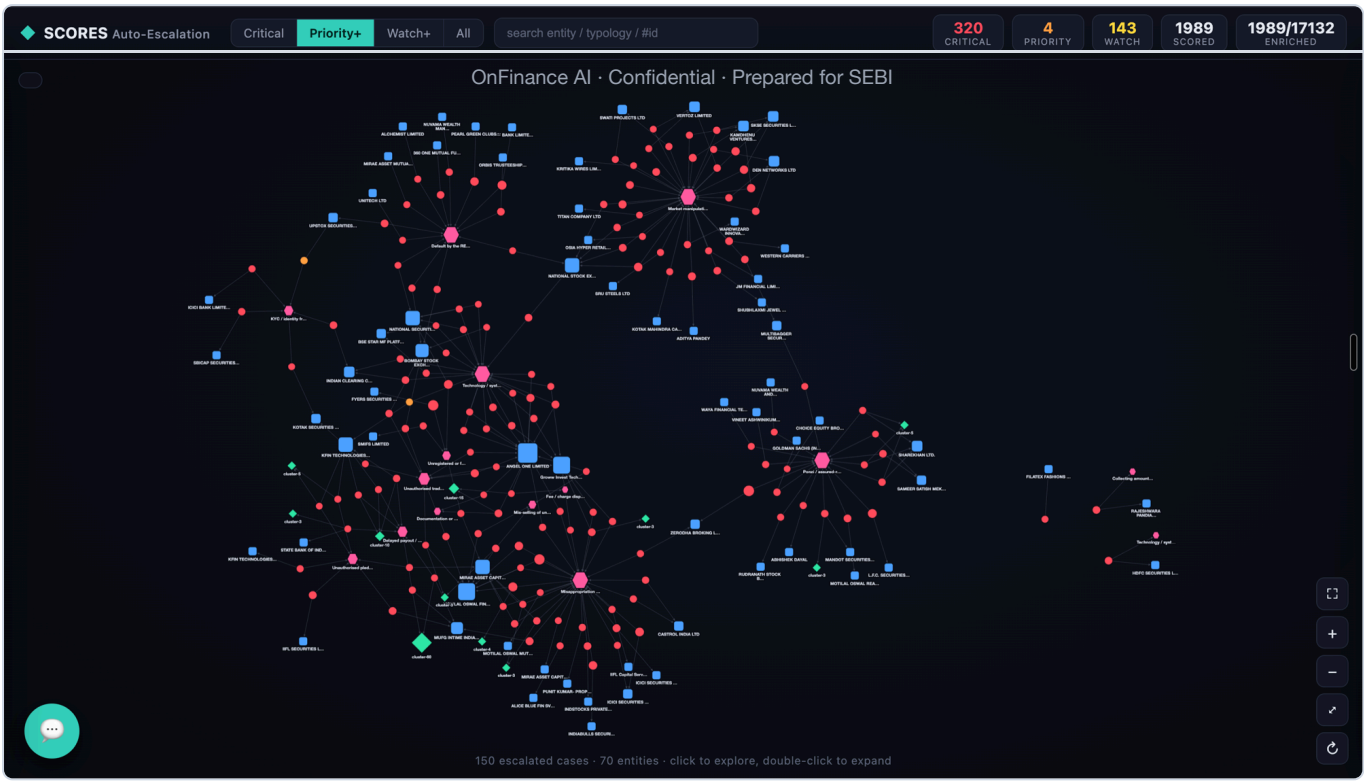


Figure 1 — The escalation knowledge graph. Red/blue nodes are complaints and entities; pink hexagons are typologies; green diamonds are consolidated clusters. Systemic hubs (e.g. “Market manipulation”, “Default by the RE”, large registrar clusters) emerge visually.

5 · The Systemic-Criticality Score (SCS)

Each complaint is scored 0–100 on eleven independent dimensions. The additive dimensions combine by weight; vulnerability (F) is a capped multiplier, not an additive term; and any single **hard trigger** overrides the maths to force Tier 3.

#	Dimension	Weight	Core question
A	Scale / Mass-Population	18%	How many investors, directly or by pattern?
B	Financial Magnitude	16%	How much money is at stake — per case and in aggregate?
C	Fraud & Severity Typology	16%	Is the alleged conduct inherently grave?
D	Entity Systemic Importance	10%	Is the RE systemically important or under stress?
E	Velocity & Emergent Patterns	10%	Is this surging or a brand-new typology?
G	Market-Integrity & Contagion	10%	Could this threaten price discovery, settlement or trigger a run?
I	Process & Resolution Failure	8%	Is the RE's response evasive or chronically failing?
J	Cross-Portal & External Correlation	6%	Does it corroborate other intelligence? *(data-dependent — see §12)*
H	Distress & Escalation Signals	3%	Does the language signal fraud, coordination or acute distress?
K	Legal & Reputational Sensitivity	3%	Does this carry institutional / legal / reputational risk for SEBI?
F	Vulnerable-Investor Impact	× multiplier	Are the affected investors disproportionately vulnerable? (amplifier, +0–30%)

The composite SCS maps to four tiers, each with its own routing and SLA:

Tier	SCS band	Meaning	Routing & SLA
Tier 0 · Routine	0–39	No systemic signal	Standard SCORES queue
Tier 1 · Watch	40–59	Warrants analyst eyes	Analytics watch-list; weekly digest
Tier 2 · Priority	60–79	Likely systemic	Division / Department Head within 48 hrs
Tier 3 · Critical	80–100 *or any hard trigger*	Confirmed / strongly-suspected systemic risk	Senior-management / WTM-level desk within 24 hrs



Figure 2 — A live case (#1741193, an assured-return / Ponzi allegation against a broker). The right panel shows the AI summary, the two hard triggers that fired, the full 11-dimension breakdown, the extracted signals and the verbatim evidence — the complete, explainable basis for its Critical rating.

6 · The Auto-Escalation Dashboard

The dashboard is the senior-management working surface. It has three coordinated panels.

- **Escalation queue (left).** Every escalated case, ranked by SCS, colour-coded by tier, with its typology and the dimension codes of the triggers that fired. One click opens the full case.
- **Knowledge-graph canvas (centre).** The live escalation graph. Click any node to focus it and its neighbours; double-click to expand; the whole graph can be maximised to full screen. This is where clusters and cross-entity patterns become obvious.
- **Analytics (right).** Three tabs — *Clusters* (the consolidated case-files, ranked), *Dimensions* (what is driving escalations), and *Entities* (the REs with the most escalations).



Figure 3 — Focus view. Selecting the "Default by the RE / broker default" typology highlights every escalated complaint alleging it — an instant, evidence-linked map of a single systemic theme across entities and states.



Figure 4 — The Clusters tab: root-cause case-files ranked by size and tier (e.g. an RTA cluster of 60 complainants across 19 states; a "templated 100%" cluster flagged as a possible coordinated filing rather than genuine mass harm).

7 · AI Case Summaries

Every case carries a one- or two-sentence, plain-English summary generated by the AI from the complaint text — *what the issue is and why it matters to a regulator*. It is generated once and cached, so it opens instantly thereafter. An officer can grasp a case before reading a single line of the raw complaint.

> **Example (case #1741193).** “The investor claims to have invested ₹32 lakh with LFS Broking (a now-banned broker) based on guaranteed monthly returns, but the company stopped payments and is withholding capital since 2022-23. This matters to SEBI because it demonstrates a clear Ponzi / guaranteed-return scheme, requires verification of Investor Protection Fund eligibility for recovery, and highlights systemic compliance failure at a licensed broker.”

8 · Ask-the-Graph — natural-language investigation

Officers do not need to learn a query language. The built-in chatbot uses **GraphRAG**: a question in plain English is converted by the AI into a single **read-only** database query, executed against the graph, and answered in words — grounded entirely in the actual results.

- **Read-only by construction.** The generated query is validated before it runs; anything that could modify the graph is rejected. The chatbot can never change data.
- **Abbreviation-aware.** It is given the real entity list, typologies and category names, so “Zerodha”, “NSE”, “Motilal” or “KFin” resolve to the correct entities automatically.
- **Auditable.** Every answer shows the exact query it ran, so the reasoning is transparent and reproducible.

Ask the graph GraphRAG · read-only

from the graph.

Top 5 entities by critical complaints

Top 5 Entities by Critical Complaints

Rank	Entity Name	Critical Count	Key Typologies
1	**Groww Invest Tech Private Limited**	31	Wrong execution, delayed payouts, technology/system failures, misappropriation, unauthorized trading, documentation deficiencies
2	**ANGEL ONE LIMITED**	29	Technology/system failures, documentation deficiencies, misappropriation, delayed payouts, unauthorized trading, mis-selling
3	**MOTILAL OSWAL MUTUAL FUND**	14	Delayed payouts, documentation deficiencies, misappropriation, unauthorized trading
4	**KFIN TECHNOLOGIES PVT LTD**	12	Delayed payouts, unauthorized pledging/securities transfer, documentation deficiencies
5	**MOTILAL OSWAL FINANCIAL SERVICES LIMITED**	11	Mis-selling to retail investors, misappropriation, unauthorized trading

Summary: Groww Invest Tech leads with 31 critical

Top 5 entities by critical complaints

How many complaints allege misappropriation?



Figure 5 — "Top 5 entities by critical complaints" answered from the live graph, with the key typologies per entity. The underlying query is shown to the officer for full transparency.

9 · Auto-Escalation & Notification Engine

This is the mechanism by which a critical issue stops being a row in a database and becomes an **alert on the right person's desk**. When the scoring engine assigns a case (or a consolidated cluster) a tier — or fires a hard trigger — the notification engine determines *who* should know, *how* they are told, and *how fast*.

9.1 The escalation cues — what causes an escalation

An escalation is raised on any of three cue types. Each is configurable by SEBI.

Cue type	What fires it	Resulting tier
Hard trigger (9 defined)	Any single grave signal — a Grave typology (misappropriation, Ponzi, manipulation, RE default, fake entity); single case ≥ ₹5 cr or misappropriation ≥ ₹2 cr; a mass cluster (≥100 complainants / ≥3 states / cross-RE product); an MII in an integrity failure; a credible market-integrity signal; a cluster of second-level reviews against one RE; senior-management allegation + grave typology.	Critical
Composite threshold	The weighted 11-dimension SCS crosses a tier band (60 → Priority, 80 → Critical).	Priority / Critical
Emergent pattern	A velocity spike (≥4× an RE/product's baseline) or a newly-formed mass cluster crossing the scale threshold.	Priority → Critical

HARD-TRIGGER OVERRIDE — any single hard trigger routes the case to Tier 3 regardless of the computed score. The composite score governs only the cases that do not fire a hard trigger. A one-victim Ponzi allegation escalates just as forcefully as a thousand-victim cluster.

9.2 Routing — who gets told

A configurable **routing matrix** maps each escalation to a recipient, a channel and an SLA, based on its tier plus any special dimension. Recipients are SEBI-maintained distribution lists / mailboxes.

Escalation	Designated recipient	Channel	SLA
Tier 3 · Critical	Senior-management / WTM-level desk + relevant department	Immediate email + dashboard alert	Within 24 hrs
Tier 2 · Priority	Division / Department Head	Daily priority digest + dashboard	Within 48 hrs
Tier 1 · Watch	Dealing officer / analytics desk	Weekly digest + watch-list	Best-effort
Special: **MII** implicated	MII supervision cell	Immediate email	Within 24 hrs
Special: **price-sensitive / confidential**	Restricted list; confidential-handling flag set	Email (restricted); hidden on general dashboard	Within 24 hrs

Special: **acute investor distress**	Investor-services / duty-of-care desk	Immediate email	Prompt, humane handling

9.3 Channels

- **Email (primary)** — a structured escalation notice to the designated mailbox/distribution list, sent via SEBI's SMTP relay.
- **In-dashboard alert feed** — the live Tier 2/3 queue is the always-on channel; new Critical items surface at the top in real time.
- **Digests** — a daily Priority digest and a weekly Watch digest, so lower tiers are seen without noise.
- **Optional webhook** — Slack / Microsoft Teams / MS-Exchange, for desks that prefer chat-based alerts.

9.4 The escalation notice — what the officer receives

Every notification is a complete, explainable case-file — not just "case X escalated". It carries:

▲ SCORES Auto-Escalation — **TIER 3 · CRITICAL** — action within 24 hrs

Case	#1741193 · ZERODHA BROKING LIMITED · Maharashtra
SCS / Tier	55.4 · Critical (+30% vulnerability multiplier)
Why (triggers)	C: Grave typology (Ponzi / assured-return) · K: senior-management allegation + grave typology
Top dimensions	Legal/Reputational 100 · Distress 100 · Fraud-Severity 98 · Financial 65
AI summary	Investor invested ₹32 lakh with a now-banned broker on guaranteed returns; payments stopped since 2022-23 — a clear Ponzi/guaranteed-return scheme; IPF-eligibility check needed.
Evidence	"…is Being Banned for 5 yrs … for assuring fixed return monthly and the MD … is jailed."
Exposure	₹32,00,000 · confidence: high · consolidated case-file: n/a (single)
Recommended	Senior-management review; verify IPF eligibility · **Open case** ▸ (deep-link to dashboard)

The recipient can open the case in one click, see the full graph neighbourhood, and **confirm, de-escalate or re-route** — every decision is logged.

9.5 Noise control — why the desk is not flooded

- **Cluster de-duplication.** A mass event escalates as *one* consolidated case-file, not N separate alerts. (In the pilot, 320 critical complaints collapse to ~21 Tier-3 case-files.)
- **Confidence gating.** Low-confidence extractions are marked "provisional — needs verification" rather than asserted, and grave hard-triggers require a minimum confidence.
- **Coordinated-campaign flag.** Near-identical (templated) filings are labelled as possible astroturfing and are *not* counted as genuine mass harm.
- **Cool-downs & idempotency.** One notice per case per state-change; a re-scored case does not re-alert unless its tier changes. Per-RE baselines stop a high-volume entity being flagged purely for its size.

9.6 Implementation status & integration

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> **What is built.** The scoring engine already produces, for every case, the tier, the firing dimensions and hard-triggers, the evidence quotes, the AI summary, the exposure estimate and a confidence level — i.e. the full notice payload. The live dashboard is the always-on escalation feed today.

To go live on email/chat, the engine emits each escalation as an event to a dispatch module (an “escalation outbox”); SEBI supplies (a) an SMTP relay or webhook endpoint and (b) the recipient mapping for the routing matrix in §9.2. Because the payload and routing are already computed, this is a thin, configuration-driven integration rather than new analytics — and the routing table, thresholds, recipients and digest cadence are all SEBI-tunable without code changes.

10 · Explainability, Governance & Guardrails

- **Explainable by default.** Every escalation carries the firing dimensions, the specific triggers, the SCS breakdown, the quoted evidence and a confidence level. No black-box alerts.
- **Human-in-the-loop.** The engine escalates and recommends; it never closes, dismisses, adjudicates or enforces. Officers confirm, de-escalate or re-route; corrections feed back as training signal.
- **False-positive control.** Cluster de-duplication, precision-guarded hard-triggers, per-RE baselines and coordinated-campaign detection keep the critical queue meaningful (pilot: ~16% of scored complaints, consolidating to a handful of case-files).
- **Read-only & private.** The platform never writes to SCORES; complainant identity is stored only as a one-way hash; price-sensitive cases inherit a confidential-handling flag with restricted visibility.
- **Auditable.** Every score, rationale and officer decision is retained for audit and periodic bias review, particularly of vulnerability-weighted decisions.

11 · Deployment & Isolation

- **Standalone service.** The entire platform is a single self-contained application plus a Neo4j graph database. It exposes its own API and dashboard.
- **Zero impact on existing SCORES.** It reads existing complaint data read-only (or via existing read APIs) and writes nothing back. No SCORES, CRUD or front-end service is modified; the platform ships its own UI on its own route.
- **VM / on-prem ready.** Containerised (Docker); the AI can be pointed at an on-prem / air-gapped LLM gateway without code changes.
- **Resumable & idempotent.** The enrichment and scoring jobs are batch, resumable and re-runnable, so the graph can be refreshed on a schedule as new complaints arrive.

12 · Roadmap

Item	Description
Full-population enrichment	Extend AI scoring from the ~2,000-case pilot to all 17,132 complaints (a background job) for statistically complete KPIs.
Live notification integration	Connect the escalation outbox to SEBI's SMTP relay / Slack / Teams and load the recipient routing matrix (§9.6).
SEBI control panel	A UI to tune thresholds, weights, typology mappings, routing and recipients — with change history — so no code change is ever needed.
Cross-portal correlation (Dimension J)	Wire in Market-Intelligence-portal / examination signals when access is granted (currently scored 0 and flagged, never fabricated).
Temporal baselines & recidivism	Rolling per-RE baselines and repeat-conduct detection as more historical data is ingested.
Feedback-driven recalibration	Use officer confirm / de-escalate decisions to periodically re-tune weights and thresholds.

13 · Annexure — Parameter Quick Reference

Composite weights

A 18% · B 16% · C 16% · D 10% · E 10% · G 10% · I 8% · J 6% · H 3% · K 3% (sum 100%); F = capped multiplier +0–30%. All configurable.

Tier bands

0–39 Routine · 40–59 Watch · 60–79 Priority · 80–100 (or any hard trigger) Critical.

Hard triggers (any one → Tier 3)

#	Dimension	Condition (default)
1	A — Scale	cluster ≥ 100 complainants, or ≥ 3 states, or cross-RE product ≥ 3 REs
2	B — Magnitude	single ≥ ₹5 cr; misappropriation ≥ ₹2 cr; cluster aggregate ≥ ₹50 cr
3	C — Typology	any “Grave” typology (misappropriation, manipulation, Ponzi, RE default, fake entity)
4–5	D — Entity	MII in an integrity/operational failure; payout/redemption-delay cluster (solvency stress)
6	G — Integrity	credible manipulation / settlement / run-risk / price-sensitive signal
7	I — Process	cluster of second-level-review escalations against one RE
8	J — Correlation	corroborates an open MI-portal subject / live examination *(when data available)*
9	K — Sensitivity	allegation against RE senior management/promoters + a High/Grave typology

End of document. All thresholds, weights, typology mappings, routing rules and recipients in this platform are configurable defaults, tunable by SEBI without code changes.